The documentation and process conversion measures necessary to comply with this revision shall be completed by 22 May 1995

INCH-POUND

MIL-S-19500/397D
22 December 1994
SUPERSEDING
MIL-S-19500/397C
16 May 1994

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON
TYPES 2N3743, 2N4930, AND 2N4931
JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by the U.S. Army Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNP, silicon, high-voltage transistor. Three levels of product assurance are provided for each device type as specified in MIL-S-19500. For JAN product assurance level (see 6.3.1). Two levels of product assurance for die (element evaluation).

1.2 Physical dimensions. See figure 1 (TO-39) and figure 2 for JANHC and JANKC (die) dimensions.

1.3 Maximum ratings.

	P_T		V_{CBO}	V_{EBO}	V_{CEO}	I_C	T_J and T_{STG}
	T_A = +25°C 1/	T_C = +25°C 2/					
	W	W	V dc	V dc	V dc	mA dc	°C
2N3743	1	5	300	5	300	200	-65 to +200
2N4930	1	5	200	5	200	200	-65 to +200
2N4931	1	5	250	5	250	200	-65 to +200

1/ Derate linearly at 5.71 mW/°C for T_A > +25°C.
2/ Derate linearly at 28.6 mW/°C for T_C > +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Research Laboratory, Electronics and Power Sources Directorate, ATTN: AMSRL-EP-RD, Fort Monmouth, NJ 07703-5601, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics.

$\lvert h_{fe} \rvert$		h_{FE1} 1/	h_{FE4} 1/	$V_{BE(sat)2}$ 1/	$V_{CE(sat)1}$ 1/	C_{obo}
I_C = 10 mA dc V_{CE} = 20 V dc f = 20 MHz		I_C = 0.1 mA dc V_{CE} = 10 V dc	I_C = 30 mA dc V_{CE} = 10 V dc	I_C = 30 mA dc I_B = 3 mA dc	I_C = 30 mA dc I_B = 3 mA dc	I_E = 0 V_{CB} = 20 V dc f ≥ .1 MHz
				V dc	V dc	pF
Min	2.0	30	50			
Max	8.0		200	1.2	1.2	15

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

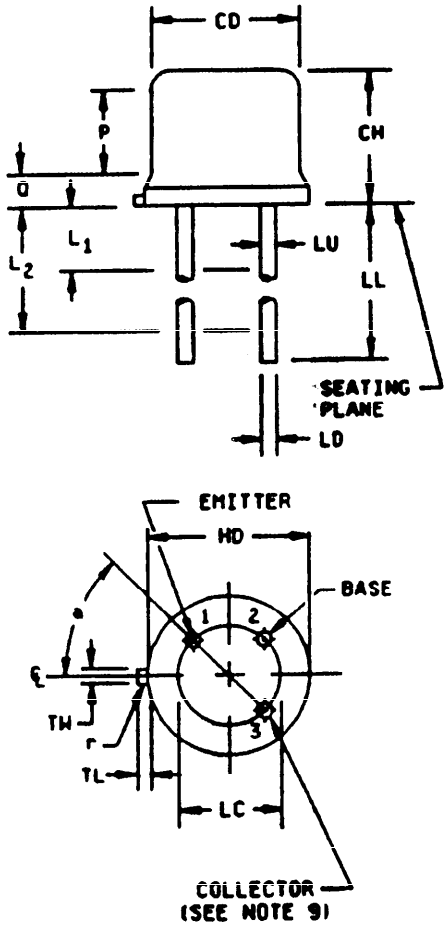
(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

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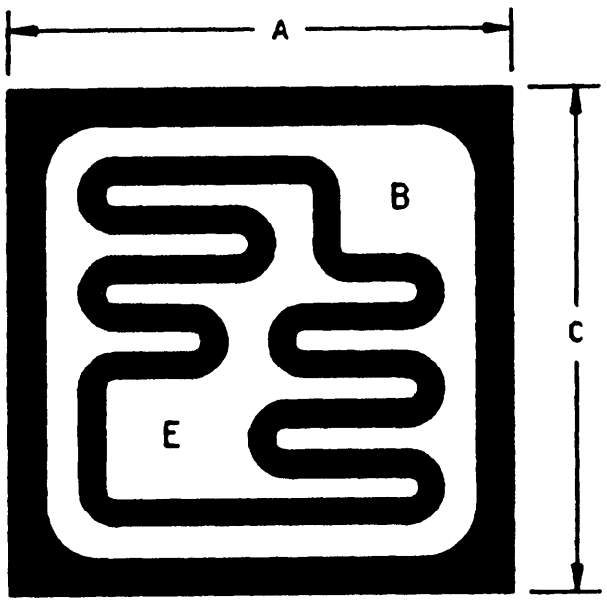


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CH	.240	.260	6.10	6.60	
LC	.200 TP		5.08 TP		7
LD	.016	.021	0.41	0.53	7,8
LU	.016	.019	0.41	0.48	8,9
HD	.335	.370	8.51	9.40	
CD	.305	.335	7.75	8.51	
TW	.028	.034	0.71	0.86	3
TL	.029	.045	0.74	1.14	5
LL	.500	.750	12.70	19.05	
L_1		.050		1.27	8,9
L_2	.250		6.35		8,9
P	.100		2.54		6
Q		.050		1.27	5
r		.010		0.25	11
a	45° TP		45° TP		7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
4. TL measured from maximum HD.
5. Outline in this zone is not controlled.
6. CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001, -.000 inch (1.37 + 0.03, -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
8. LU applies between L_1 and L_2. LD applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
9. All three leads.
10. The collector shall be electrically and mechanically connected to the case.
11. r (radius) applies to both inside corners of tab.
12. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (TO-39).

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A version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.041	.041	1.04	1.04
C	.041	.041	1.04	1.04

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:
   Thickness: .006 inch (0.15 mm) to .012 inch (0.30 mm).
   Top metal: Aluminum 17,500 Å minimum, 20,000 Å nominal.
   Back metal: Gold 2,500 Å minimum, 3,000 Å nominal.
   Back side: Collector.
   Bonding pad: B = .004 inch (0.10 mm) x .005 inch (0.13 mm).
   E = .004 inch (0.10 mm) x .0055 inch (0.14 mm).

FIGURE 2. JANHC and JANKC (A-version) die dimensions.

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3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figure 1 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be gold, tin or solder. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-S-19500 and table II herein.

4.2.2 JANHC and JANKC devices. Qualification for devices shall be in accordance with appendix H of MIL-S-19500.

4.3 Screening (JANS, JANTX, AND JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{CBO1}	Not applicable
11	I_{CBO1} and h_{FE4} ΔI_{CBO} = 100 percent of initial value or 50 nA, whichever is greater.	I_{CBO1} and h_{FE4}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO1} = 100 percent of initial value or 50 nA dc, whichever is greater; Δh_{FE4} = ±15 percent.	Subgroup 2 of table I herein; ΔI_{CBO1} = 100 percent of initial value or 50 nA dc, whichever is greater; Δh_{FE4} = ±20 percent.

4.3.1 Power burn-in test conditions. Power burn-in conditions are as follows:

$V_{CB} \geq 30$ V dc
P_T = 1 W at T_A = +30°C ±5°C.

NOTE: No heatsink or forced air cooling on the devices shall be permitted. For JANS, P_T adjusted to achieve T_J of +187.5°C ±12.5°C.

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4.3.2 Screening JANHC or JANKC. Screening of die shall be in accordance with appendix H of MIL-S-19500.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables IVa (JANS) and IVb (JANTX and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 30 V dc; P_T adjusted to achieve $\Delta T_J \geq +100°C$. No heat sink on the devices shall be permitted.
B5	1027	V_{CB} = 30 V dc, P_T adjusted to achieve T_J and time required in MIL-S-19500.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B3	1027	V_{CB} = 30 V dc; P_T = 1 W at T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. No heat sink on the devices shall be permitted.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.3.1 Group C inspection, (table V of MIL-S-19500).

Subgroup	Method	Condition
C2	2036	Test condition E.
C6	1026	V_{CB} = 30 V; P_T = 1 W at T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. No heat sink or forced air cooling on the devices shall be permitted.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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4.5.2 Thermal resistance (to be performed for qualification inspection only). Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. I_M: Measurement current . . . . . 10 mA.

b. V_{CE}: Measurements current (same as V_H). . . . . 20 V.

c. I_H: Collector heating current . . . . . 0.15 A dc.

d. V_H: Reference point temperature shall be +25°C ≤ T_R ≤ +35°C and recorded before the test is started.

e. Mounting arrangements shall be with heat sink to case.

f. Maximum limit shall be $R_{\theta JC}$ = 35°C/W.

4.5.2.1 Thermal resistance for JANS. Thermal resistance measurements shall be performed in accordance with MIL-STD-750, method 3131. The maximum limit of $\Theta_{JC(max)}$ shall be 35°C/W. The following test conditions shall apply to both group B and screening step 3C, for screening step 3C, tH shall be 10 ms minimum:

a. I_M: Measurement current . . . . . 10 mA.

b. V_{CE}: Measurements current (same as V_H) . . . . . 25 V.

c. I_H: Collector heating current . . . . . 0.2 A.

d. V_H: Collector-emitter heating voltage . . . . . 25 V.

e. t_H: Heating time . . . . . 1 second minimum.

f. t_{MD}: Measurement delay time . . . . . 50 µs.

g. t_{SW}: Sampling window time . . . . . 10 µs maximum.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage, collector to base	3001	Bias condition D, I_C = 100 µA dc	$V_{(BR)CBO}$			
2N3743				300		V dc
2N4930				200		V dc
2N4931				250		V dc
Breakdown voltage, collector to emitter	3011	Pulsed (see 4.5.1), bias condition D, I_C = 1.0 mA dc	$V_{(BR)CEO}$			
2N3743				300		V dc
2N4930				200		V dc
2N4931				250		V dc
Breakdown voltage, emitter to base	3026	Bias condition D, I_E = 100 µA dc	$V_{(BR)EBO}$	5		V dc
Collector to base cutoff current	3036	Bias condition D, I_E = 0	I_{CBO1}		250	nA
2N3743		V_{CB} = 250 V dc				
2N4930		V_{CB} = 150 V dc				
2N4931		V_{CB} = 200 V dc				
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 4 V dc	I_{EBO}		150	nA dc
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), I_C = 0.1 mA dc, V_{CE} = 10 V dc	h_{FE1}	30		
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), I_C = 1.0 mA dc, V_{CE} = 10 V dc	h_{FE2}	40		
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), I_C = 10 mA dc, V_{CE} = 10 V dc	h_{FE3}	40		
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), I_C = 30 mA dc, V_{CE} = 10 V dc	h_{FE4}	50	200	

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), I_C = 50 mA dc, V_{CE} = 20 V dc	h_{FE5}	30		
Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1), I_C = 30 mA dc, I_B = 3 mA dc	$V_{CE(sat)1}$		1.2	V dc
Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1), I_C = 10 mA dc, I_B = 1 mA dc	$V_{CE(sat)2}$		1.0	V dc
Base emitter voltage (saturated)	3066	Test condition A, pulsed (see 4.5.1), I_C = 10 mA dc, I_B = 1 mA dc	$V_{BE(sat)1}$		1.0	V dc
Base emitter voltage (saturated)	3066	Test condition A, pulsed (see 4.5.1), I_C = 30 mA dc, I_B = 3 mA dc	$V_{BE(sat)2}$		1.2	V dc
Subgroup 3						
High temperature operation:		T_A = +125°C				
Collector to base cutoff current	3036	Bias condition D	I_{CBO2}		100	µA dc
2N3743		V_{CB} = 250 V dc				
2N4930		V_{CB} = 150 V dc				
2N4931		V_{CB} = 200 V dc				
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), V_{CE} = 10 V dc, I_C = 30 mA dc	h_{FE6}	25		
Subgroup 4						
Open circuit (output capacitance)	3236	V_{CB} = 20 V dc, I_E = 0, f ≥ 0.1 MHz	C_{obo}		15	pF
Input capacitance (output open circuited)	3240	V_{EB} = 1 V dc, I_C = 0, f ≥ 0.1 MHz	C_{ibo}		400	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Small-signal current gain	3306	V_{CE} = 20 V dc, I_C = 10 mA dc, f = 20 MHz	$\|h_{fe}\|$	2	8	
Small-signal current gain	3206	V_{CE} = 10 V dc, I_C = 10 mA dc, f = 1 kHz	h_{fe}	30	300	
Subgroup 5						
Safe operating area (dc operation)	3051	T_C = +25°C, t ≥ 1 second, 1 cycle				
Test 1		I_C = 50 mA dc, V_{CE} = 20 V dc				
Test 2		I_C = 10 mA dc, V_{CE} = 100 V dc				
Test 3						
2N3743		I_C = 3.3 mA dc, V_{CE} = 300 V dc				
2N4930		I_C = 5 mA dc, V_{CE} = 200 V dc				
2N4931		I_C = 4 mA dc, V_{CE} = 250 V dc				
Electrical measurements		See table III, steps 1 and 3				

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification conformance inspection
	Method	Conditions	
Subgroups 1, 2, and 3			
Not applicable			
Subgroup 4			22 devices, c = 0
Thermal resistance	3131	See 4.5.2	
Subgroup 5			5 devices, c = 0
Barometric pressure (2N3743 and 2N4931 only)	1001	Pressure = 8 mmHg, normal mounting, t = 60 seconds, minimum	

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead material and finish as required (see 3.3.1).

c. Product assurance and type designation, and for die acquisition the JANHC and JANKC identification (see figure 2 and 6.4).

6.3 Substitution information. Devices from this specification supersede the following commercial Part or Identifying Number (PIN). This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

Military PIN	Superseded commercial types
2N3743	SUN1446H, SS4238H
2N4930	SUN1446H, SS5152H
2N4931	SUN1446H, ST1390H ST147H

6.3.1 Replacement data. JANTX devices are direct replacements for JAN devices (example, JANTX2N3743 for JAN2N3743).

6.4 Suppliers of JANHC and JANKC die. The qualified die suppliers with the applicable letter version (example, JANHCA2N3743) will be identified on the qualified products list.

JANC ordering information		
PIN	Manufactures	
	33178	
2N3743	JANHCA2N3743 JANKCA2N3743	
2N4930	JANHCA2N4930 JANKCA2N4930	
2N4931	JANHCA2N4931 JANKCA2N4931	

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodian:
Army - ER
Air Force - 17

Review activities
Air Force - 85, 99

Preparing activity:
Army - ER

Agent:
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/397D	2. DOCUMENT DATE (YYMMDD) 941222

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, TYPES 2N3743, 2N4930, AND 2N4931, JANT, JANTXV AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
DIRECTOR, US ARMY RESEARCH LABORATORY	(1) Commercial 908-544-3450	(2) AUTOVON 995-3450
c. ADDRESS (Include Zip Code) POWER SOURCES DIRECTORATE ATTN: AMSRL-EP-RD FORT MONMOUTH, NJ 07703-5601	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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